Project Idea and Background

For my semester-long design project, I would like to build an inventory and order management system for Lylac Jewels, which is a real jewelry business owned by my sister. The system would help the business owner manage jewelry products, available quantities, prices, customer orders, and order statuses. The owner would have a secure login and dashboard where she could add new products, update product details, change inventory quantities, and mark items as available or sold out. The system would also allow her to view orders and update their status as pending, confirmed, shipped, delivered, or canceled. If time allows, I would also like to include low-stock notifications and basic sales reports.

I chose this project because the Lylac Jewels website is currently being developed, and this system could eventually become part of the complete website. My sister could act as the client by explaining what the business needs and giving me feedback during development. This would give me experience gathering requirements from a real user and making changes based on that feedback. I also hope to improve my programming, database design, testing, and web-development skills. The completed system would be useful for the business and could also be included in my portfolio or resume.

My programming background includes basic experience with Python, databases, SQL, and GitHub. I have worked on smaller Python assignments and have studied topics such as database tables, primary and foreign keys, normalization, and software-development processes. I am still developing my web-programming skills, so this project would give me practical experience connecting a user interface to a database and building a complete application.

I would like to use Python with Flask for the back end, HTML and CSS for the website pages, and SQLite or MySQL for storing product and order information. I may use JavaScript for interactive features and GitHub for version control. I would prefer to avoid mobile-app development and complicated online payment processing during this project because I want to focus on completing the inventory and order-management features correctly.

I plan to complete this project individually, so I do not have any preferred group members. Since the project is connected to a real business, I will be able to test it with realistic products and receive regular feedback from the business owner. My main goal is to create a working and easy-to-use system that can later be connected to the full Lylac Jewels website.